MIL-P-28941(MC)
8 January 1970
SUPERSEDING
See Section 6.

MILITARY SPECIFICATION

PUBLIC ADDRESS SET: HAND HELD, AN/PIQ-5( )

1. SCOPE

1.1 This specification covers the requirements for one type of battery operated hand held public address set.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-485	-	Enamel, Semi-gloss, Rust-inhibiting.

MILITARY

MIL-T-152	-	Treatment, Moisture And Fungus-resistant, Of Communications, Electronic, And Associated Electrical Equipment.
MIL-T-704	-	Treatment And Painting Of Materiel.
MIL-P-15024	-	Plate, Identification-information And Marking For Identification of Electrical, Electronic And Mechanical Equipment.
MIL-E-16400	-	Electronic Equipment, Naval Ship And Shore, General Specification.
MIL-E-17555	-	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of.
MIL-E-21981	-	Electronics Type Designations, Identification Plates And Markings, Requirements For.

FSC 5830

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MIL-R-22732 - Reliability Requirements For Shipboard And Ground Electronic Equipment.
MIL-P-55110 - Printed Wiring Boards.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-108 - Definition Of And Basic Requirement For Enclosure For Electric And Electronic Equipment.
MIL-STD-129 - Marking For Shipment And Storage.
MIL-STD-130 - Identification Marking Of U.S. Military Property.
MIL-STD-242 - Electronic Equipment Parts, Selected Standards.
MIL-STD-454 - Standard General Requirements For Electronic Equipment.
MIL-STD-810 - Environmental Test Methods.

PUBLICATIONS

TM-02543B-15 - U.S. Marine Corps Technical Manual, Public Address System, AN/PIQ-5A.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Prior to the submission of any complete production units for acceptance, the contractor shall furnish two first articles for testing. First article testing shall be under the direction and supervision of the cognizant Government inspector to assure conformance to this specification. Examination and testing shall be as specified herein. Approval of the first article by the cognizant Government inspector shall not relieve the contractor of the responsibility to

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supply equipment conforming to this specification. Any changes or deviations of the production items from the approved first article shall be approved by the contracting officer. The first articles, upon refurbishing, may be included in the final shipment as part of the total quantity to be furnished under the contract. First articles are defined as an identical sample of production items to be subsequently produced in fulfillment of the contract.

3.2 Parts and materials. The selection of parts and materials listed in MIL-STD-242 are preferred. In all cases, the general material requirements as to flammability, fungus-resistance and nontoxicity as required by MIL-E-16400 shall be met.

3.2.1 Fungus treatment. The public address set shall be fungus-proofed by the selection of parts and materials which are nonnutrient for fungus, or the parts and materials shall be treated in accordance with MIL-T-152 prior to use in the equipment.

3.3 Design and construction. The design and construction shall conform as closely as possible to the model(s) (see 3.10) provided and TM-02543B-15. The public address set shall be a lightweight, portable unit capable of amplifying the human voice with sufficient magnitude so that instructions and commands may be understood over distance too great for normal voice communications. The complete public address set shall consist of an amplifier-speaker assembly contained in a single unit with a pistol grip handle, designed so that it can be lifted and held by one hand. The unit shall be completely self contained, including the battery power source, controlled by an OFF-ON switch located in the pistol grip handle. A water-tight transit case shall be provided for housing the amplifier-speaker assembly and all accessories. Accessories shall consist of the following:

QTY	ITEM
1	Carrying Case
1	Adjustable Carrying Strap
1	Microphone

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QTY	ITEM
1	25-foot microphone extension cord with connectors
1	Multi-unit adapter 1/

1/ This accessory provided only if specified in the contract or order (see 6.2).

3.3.1 Printed circuits. Printed circuits shall be in accordance with MIL-P-55110.

3.3.2 Amplifier housing. The housing for the amplifier shall be a drawn aluminum case with a one-piece cover held in place with two spring type catches. All openings into the housing shall have gaskets to limit the entrance of water.

3.3.3 Power requirements. The equipment shall be powered by not more than ten military batteries, type BA30 or BA2030. The batteries shall be contained within the speaker-amplifier unit. No separate or external battery case shall be used. Maximum battery drain shall be 100 milliamperes when the equipment is on and no signal is applied from the microphone, and 1.75 amperes at full power output. There shall be no battery drain when the equipment is turned off.

3.3.4 Electrical characteristics. The amplifier-speaker unit shall exhibit the following electrical characteristics:

(a) Power output no less than 15 watts with no more than 10 percent distortion when modulated by a 1,000-hertz signal.

(b) Input impedance shall be 400 ohms (nominal).

(c) Output impedance shall be 10 ohms (nominal).

(d) Amplifier gain shall be 46 decibels (dB) +2 -0dB.

3.3.5 ON-OFF switches. Two parallel ON-OFF switches shall be provided. One trigger type shall be mounted in the handle. The other shall be of the push button type, and shall be mounted on the microphone. Microswitches shall be used in both the handle and the microphone.

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3.3.6 Volume control. The volume control shall be mounted within the housing of the microphone.

3.3.7 Speaker. The speaker shall be of the reentrant horn type.

3.3.8 Carrying handle. A pistol type handle shall be provided, mounted under the center of balance of the speaker-amplifier unit. The base of the handle shall be drilled and tapped for a 1/4-20 bolt so that the unit may be mounted on a standard camera type tripod.

3.3.9 Microphone. The microphone shall be of the magnetic or dynamic noise cancelling type. The microphone shall not be permanently attached to the speaker amplifier but shall be capable of attachment to the assembly by means of a clip or other suitable fasteners so that the unit can be employed as a completely self-contained amplifying megaphone. The microphone shall also be capable of rapid detachment for use as a separate hand held microphone.

3.3.9.1 Microphone cord. The microphone cord shall be permanently attached to the microphone at one end, with a connector to mate with the amplifier unit at the opposite end. The cord shall be of the spring coil type with an extended length of 6 feet.

3.3.9.2 Microphone extension cord. The microphone, when connected to the amplifier through two 25-foot extension cords, shall obtain full output power from the amplifier when modulating the microphone with the normal speaking voice.

3.3.9.3 Multi-unit adapter. When adapter is provided, it shall consist of the necessary components required to enable the use of one microphone to drive two public address sets. The adapter shall be capable of use with or without the 25-foot extension cord.

3.3.10 Carrying case. The public address set shall be provided with a metal or fiber glass carrying case. The case shall be large enough to contain the speaker-amplifier assembly and the accessories listed in 3.3 but shall not

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exceed a physical size of 18 inches by 12 inches by 15 inches overall. The cover of the carrying case shall be provided with quick release type fasteners. A handle shall be provided on the top of the carrying case. With cover in place, the case shall not leak when tested in accordance with MIL-STD-108. The case shall be chemically resistant to salt fog spray.

3.3.11 Interchangeability. Replaceable parts used in the manufacture of the item described herein shall be interchangeable with the model(s) provided (see 3.10).

3.4 Reliability. Reliability shall receive major emphasis during the design, fabrication and construction of this equipment. This emphasis shall be reflected in the assembly processes and the determination of operational techniques. The contractor shall be required to demonstrate that the system is capable of a calculated mean time between failure (MTBF) of not less than 1,500 hours and an actual MTBF of not less than 500 hours in accordance with test level number 2 of MIL-R-22732.

3.5 Dimensions. The speaker-amplifier assembly with microphone attached shall not exceed the following dimensions:

Length, 15 inches overall.

Width, 9 inches.

Height, from the top of the horn to the bottom of the handle, 13 inches.

3.6 Weight.

3.6.1 Speaker-amplifier. The weight of the speaker-amplifier assembly complete with batteries, microphone and carrying strap shall not exceed 9 pounds.

3.6.2 Adapter. The weight of the multi-unit adapter shall not exceed 5 pounds.

3.6.3 Extension cord. The weight of the 25-foot microphone extension cord shall not exceed 2 pounds.

3.7 Treatment and painting. Treatment and painting of the public address set and components, where paint does

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not effect the electrical characteristics, shall be in accordance with MIL-T-704, with paint conforming to TT-E-485. Color shall match Marine Corps green, No. 23 (see 6.3). If a fiber glass carrying case is provided, a color impregnated material matching Marine Corps green, No. 23, shall be utilized.

3.8 Identification plates. Identification plates shall be in accordance with MIL-P-15024 and shall be of aluminum alloy where practicable. Drawings of proposed identification plates shall be submitted for approval prior to the commencement of first article testing.

3.8.1 Identification markings. Identification marking of subassemblies shall be in accordance with MIL-STD-130.

3.9 Nomenclature. Nomenclature shall be requested in accordance with MIL-E-21981 and shall be submitted prior to the commencement of first article testing.

3.10 Government loaned property. The Government will provide public address sets as specified in the contract or order (see 6.2) for use during the life of the contract.

3.11 Workmanship. The public address set covered by this specification shall be manufactured and finished in accordance with MIL-E-16400, utilizing MIL-STD-454 as a guide.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.2.1).

(b) Sample (see 4.2.2).

(c) Production (see 4.2.3).

4.2.1 First article. First articles shall be subjected to tests indicated for first article in Table I. First article testing shall be witnessed by a representative of the contracting officer. Deficiencies found as a result of these tests shall be corrected before delivery of the first unit.

4.2.2 Sample. A random sample shall be selected from each lot by the Government inspector and shall be subjected to the sample tests indicated in Table I to demonstrate that the sample submitted for approval meets the requirements of this specification, with lot acceptance in accordance with MIL-STD-105.

4.2.2.1 Sampling plan. Sampling plan quantity shall be selected as follows:

Numbers of units in lot	Number of units in sample
8 and under	2
9 to 15	3
16 to 25	5
26 to 50	8
51 to 90	13
91 to 150	20
151 to 280	32
281 to 500	50
501 to 1200	80

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Table I - Tests

Reqt Para		Standard Conditions	Watertightness	Vibration	Shock	Temperature Shock	Dust	Humidity	Salt-Fog
	Performance	FA S	FA S						
3.3.12	Interchangeability	FA S P							
3.3.3	Power Requirements	FA S P		FA	FA	FA S	FA	FA	FA S
3.6.1 3.6.2 3.6.3	Weight	FA S							
3.3.3	Battery Drain	FA S P		FA S	FA	FA	FA	FA S	FA
3.3.4	Power Output	FA S P		FA S	FA	FA	FA	FA S	FA S
3.3.4	Input Impedance	FA S P							
3.3.4	Output Impedance	FA S P							
3.3.4	Amplifier Gain	FA S P							

Note: FA - First Article S - Sample P - Production

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4.2.3 Production. All public address sets shall be subjected to tests indicated for production in Table I.

4.3 Tests. The detailed procedures for conducting tests shall be prepared by the contractor and submitted to the contracting officer for review and approval 45 days prior to the initiation of any official testing. The following conditions shall be used as the basis for establishing normal requirements for conducting laboratory and bench test of the equipment:

(a) Room ambient temperature range, +20°C to +40°C.

(b) Altitude, normal ground

(c) Room ambient humidity up to 95 percent relative.

4.3.1 Vibration. Each public address set to be subjected to the vibration test shall be stowed in its carrying case, complete with accessories and rigidly mounted to the vibration table in the horizontal plane. Vibration test shall be in accordance with Method 514 of MIL-STD-810, equipment class 7, equipment mounting method A.

4.3.2 Shock. Each complete public address set to be subjected to the shock test shall be stowed in its carrying case and tested in accordance with Method 516, Procedure III of MIL-STD-810.

4.3.3 Temperature shock. The public address set shall be subjected to temperature shock test conforming to Method 503 of MIL-STD-810 except that the upper temperature shall be reduced to 65°C.

4.3.4 Dust. The public address set shall be subjected to dust test conforming to Method 510 of MIL-STD-810, equipment Class 1.

4.3.5 Humidity. The public address set shall be subjected to humidity test conforming to Method 507 of MIL-STD-810.

4.3.6 Salt fog. The public address set shall be subjected to salt fog test conforming to Method 509 of MIL-STD-810. After removal from the test chamber, the

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electrical performance of the equipment shall not be degraded more than 15 percent. When the salt accumulation has been removed by washing or hosing with fresh water, the electrical performance shall meet the requirements specified in Section 3 herein.

4.4 Test reports. The contractor shall prepare and submit to the contracting officer copies of all test reports, in duplicate, not more than 30 days after completion of the tests.

4.5 Approval. Approval of the first article shall be by the contracting officer upon satisfactory completion of all tests.

4.6 Acceptance. With exception of the first article, acceptance of all items shall be by the cognizant Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of the public address set shall be preserved, packaged and packed for the level of protection specified in the contract or order in accordance with MIL-E-17555.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The public address set is intended to provide for passing information and providing direction for formations, small crowds and other functions for which a lightweight, low powered public address set is required.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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6.2.1 Procurement requirements.

(a) Title, number and date of this specification.

(b) When multi-unit adapter is required (see 3.3).

(c) Number of public address sets to be loaned by Government (see 3.10).

(d) Levels of preservation, packaging and packing (see 5.1).

6.2.2 Contract data requirements. Data conforming to Data Item Descriptions C-03, E-14, P-01, P-07, R-02, R-05, R-07, T-01, T-02, T-03, T-04, T-05, T-10 and V-11 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from the Commandant of the Marine Corps (Code CSY-10), Headquarters U.S. Marine Corps, Washington, D. C. 20380.

6.4 First article. The first article shall be tested in accordance with Section 4.

6.5 Supersession data. This specification includes the requirements of Marine Corps Purchase Description 59-10, Hand Held Public Address Systems dated 22 June 1959.

Preparing activity:
Navy - MC
Project No. 5830-N013

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DEPARTMENT OF THE NAVY

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Commandant of the Marine Corps (CSY-10)
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Washington, D. C. 20380

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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